

CompTIA A+ Core 1 Playbook

IT Labs

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6.2.7 Configure IP Addresses

You work as the IT administrator for a small corporate network. You need to configure the workstation in the executive office so that it can connect to the local network and the internet. The workstation has two network interface cards (named Ethernet and Ethernet 2). Having two network cards allows the workstation to connect to the local network (as shown in the exhibits) and another small network, which is not yet built.

In this lab, your task is to:

- Configure IP version 4 TCP/IP using the settings specified in the table below (for both network cards).
- From the Exec computer, ping the preferred DNS server assigned to the Ethernet NIC to verify that it can communicate successfully.

TCP/IP Setting or Info	Ethernet	Ethernet 2
Subnet	192.168.0.0/24	10.0.0.0/16
IP address	Use the last valid address on the subnet.	Use the last valid address on the subnet.
Subnet mask	Use the default mask that matches	Use the default mask that

	the /24 subnet.	matches the /16 subnet.
Default gateway	Choose the appropriate address for the router as shown in the exhibits.	Leave blank
Preferred DNS server address	Use the address of an external DNS server as shown in the exhibits.	Leave blank
Alternate DNS server address	Use the address of an external DNS server as shown in the exhibits.	Leave blank

To access IPv4 settings:

1. From the Settings app, go to **Network & internet**, then **Advanced network settings**.
2. Select **More network adapter options**.
3. Right-click a connection and select **Properties**.
4. Select Internet Protocol Version 4, then select **Properties**.

Explanation

Complete this lab as follows:

1. Access the properties for the NIC named Ethernet.
 1. Right-click **Start** and then select **Settings**.
 2. Select **Network & internet**.
 3. From the right pane, select **Ethernet**.
2. Configure the IP version 4 TCP/IP settings for the Ethernet NIC.
 1. For *IP assignment*, select **Edit**.
 2. Set the dropdown to **Manual**.
 3. Toggle the switch for **IPv4**.

4. Configure the IPv4 settings as follows:
 - IP address: **192.168.0.254**
 - Subnet mask: **255.255.255.0**
 - Default gateway: **192.168.0.5**
 - Preferred DNS: **163.128.78.93**
 - Alternate DNS: **163.128.80.93**
5. Select **Save**.
3. Configure the IP version 4 TCP/IP settings for the Ethernet 2 NIC.
 1. From the left pane of the Settings app, select **Network & internet**.
 2. From the right pane, select **Advanced network settings**.
 3. Select **More network adapter options**.
 4. From the Network Connections window, right-click **Ethernet 2** and then select **Properties**.
 5. Select **Internet Protocol Version 4 (TCP/IPv4)**.
 6. Select **Properties**.
 7. Select **Use the following IP address**.
 8. Configure the Internet Protocol information as follows:
 - IP address: **10.0.255.254**
 - Subnet mask: **255.255.0.0**
 - Default gateway: **Blank**
 - Preferred DNS server: **Blank**
 9. Select **OK**.
 10. Select **Close**.
4. Ping the preferred DNS server assigned to the Ethernet NIC.
 1. Right-click **Start** and select **Terminal (Admin)**.
 2. From the PowerShell prompt, type one of the following:
 - **ping 163.128.78.93**
 - **ping 163.128.80.93**
 3. Press **Enter**.